

CMOS VLSI Design Project

Design and Analysis of a 2-Input CMOS OR Gate

Using Cadence Virtuoso

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1 Introduction

The objective of this project is to design, implement, and verify a two-input CMOS OR gate using the Cadence Virtuoso design environment. The project covers the major stages of a custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout performance analysis.

A two-input OR gate produces a logic HIGH output whenever one or both of its inputs are HIGH. The output becomes LOW only when both input signals are LOW.

The Boolean expression of a two-input OR gate is

$$Y = A + B \quad (1)$$

where A and B are the input signals and Y is the output.

In CMOS implementation, the OR function can conveniently be obtained using a two-input NOR gate followed by an inverter.

2 Objectives

The main objectives of this project are:

- To design a two-input CMOS OR gate using Cadence Virtuoso.
- To create and verify the schematic of the OR gate.
- To generate a reusable symbol for the designed circuit.
- To design the physical layout of the OR gate.
- To verify the layout using Design Rule Check (DRC).
- To verify electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform RC parasitic extraction using RCX.
- To perform post-layout simulation using the extracted circuit.
- To calculate the average power consumption of the OR gate.
- To calculate the propagation delay of the OR gate.

3 Theory of OR Gate

A two-input OR gate performs the Boolean operation

$$Y = A + B. \quad (2)$$

The output becomes logic HIGH if at least one input is HIGH.

The truth table of the two-input OR gate is shown in Table 1.

Table 1: Truth Table of a Two-Input OR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

In this project, the OR function is implemented using a NOR gate followed by an inverter. For the NOR stage,

$$X = \overline{A + B}. \quad (3)$$

The NOR output is then applied to an inverter:

$$Y = \overline{X}. \quad (4)$$

Substituting the value of X ,

$$Y = \overline{\overline{A + B}}. \quad (5)$$

Therefore,

$$\boxed{Y = A + B} \quad (6)$$

which is the required OR operation.

4 Circuit Design

4.1 Schematic Design

The two-input CMOS OR gate was designed in Cadence Virtuoso. The circuit consists of a two-input NOR gate followed by a CMOS inverter.

The two input terminals are represented by A and B , while the final output terminal is represented by Y .

The NOR gate generates

$$X = \overline{A + B}. \quad (7)$$

The inverter then complements this output to obtain

$$Y = A + B. \quad (8)$$

The schematic of the designed circuit is shown in Figure 1.

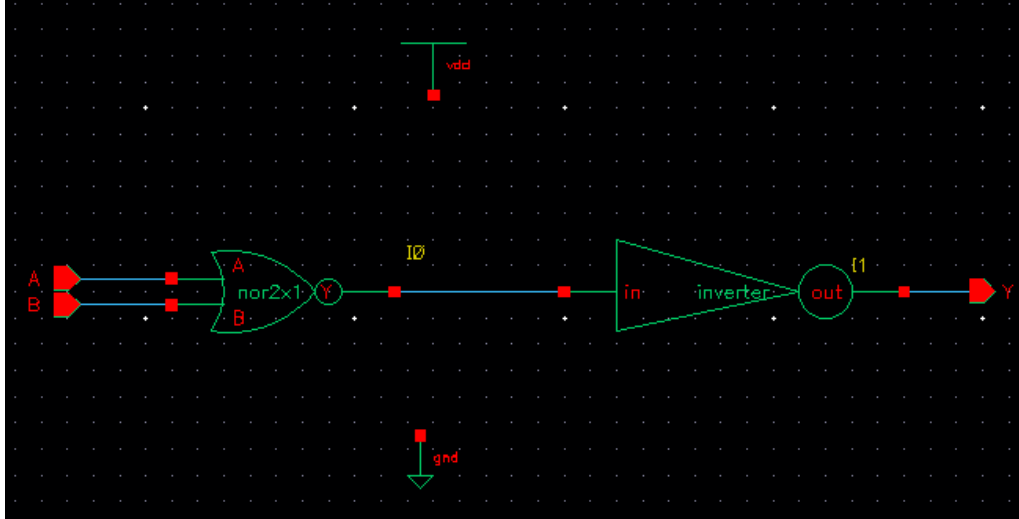


Figure 1: Schematic of the two-input CMOS OR gate in Cadence Virtuoso.

The logical operation of the circuit can therefore be expressed as

$$A, B \rightarrow \overline{A + B} \rightarrow A + B. \quad (9)$$

4.2 OR Gate Symbol

After completing and verifying the schematic, a symbol was generated for the OR gate.

The symbol contains two input terminals, A and B , and one output terminal, Y .

The generated symbol is shown in Figure 2.

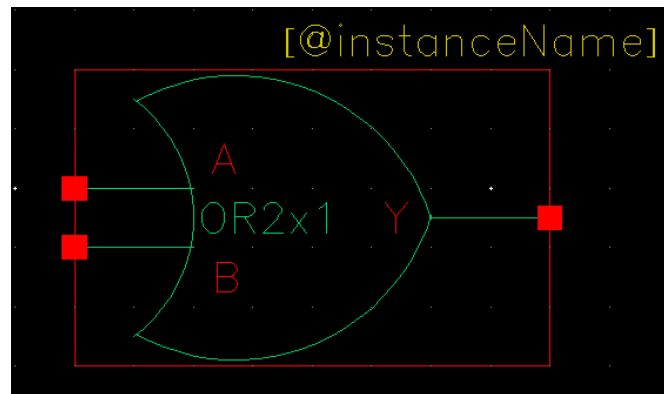


Figure 2: Generated symbol of the two-input OR gate.

The symbol allows the designed OR gate to be used as a reusable hierarchical block in more complex digital circuits.

5 Physical Layout Design

After completing the schematic and symbol, the physical layout of the OR gate was designed using Cadence Virtuoso Layout Editor.

The layout represents the physical implementation of the MOS transistors and their electrical connections.

During layout design, different fabrication layers were used, including:

- diffusion or active layers,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,
- power rail, and
- ground rail.

The PMOS transistors were placed in the upper portion of the layout, while the NMOS transistors were placed in the lower portion.

The routing was completed carefully so that the physical connectivity matched the original schematic.

The completed layout is shown in Figure 3.

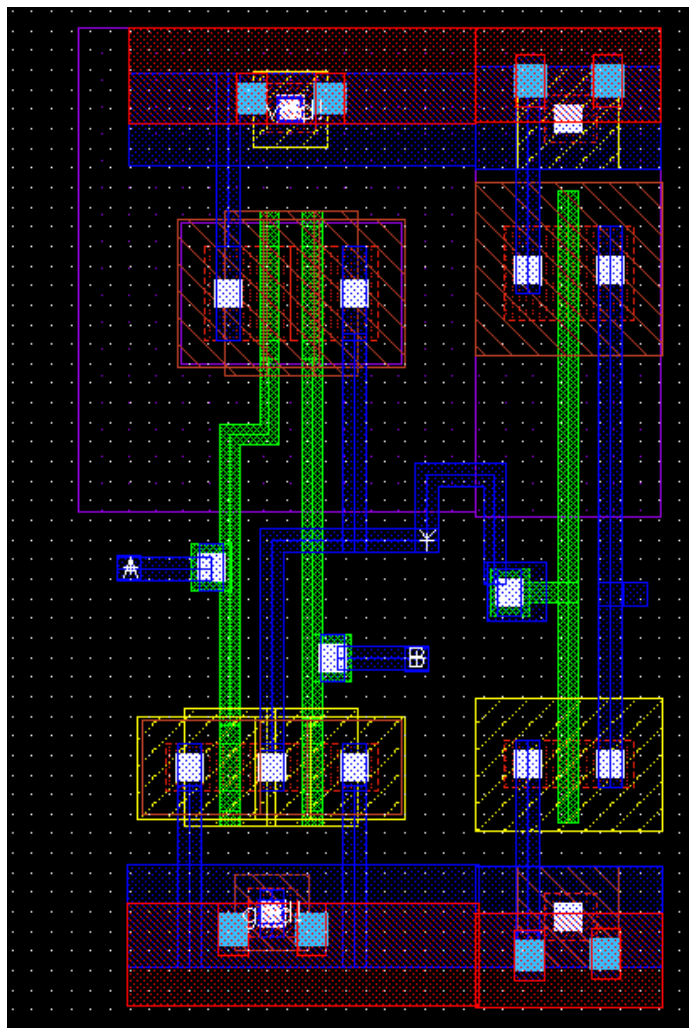


Figure 3: Physical layout of the two-input CMOS OR gate.

The layout was designed while maintaining the required minimum spacing, width, enclosure, and connectivity rules of the selected CMOS technology.

6 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,
- diffusion spacing,

- contact enclosure,
- via enclosure,
- well spacing, and
- active-area requirements.

The completed OR gate layout was checked using the DRC verification tool.

The layout successfully passed DRC.

DRC Result: PASSED

The physical layout satisfies the required fabrication design rules.

Therefore, no design-rule violation remained in the final layout.

7 Layout Versus Schematic (LVS)

After successfully passing DRC, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies different circuit properties such as:

- transistor type,
- number of transistors,
- gate connectivity,
- source and drain connectivity,
- input connections,
- output connections,
- power connections,
- ground connections, and
- overall circuit topology.

The OR gate layout successfully passed LVS verification.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that no unintended connectivity error was introduced while designing the physical layout.

8 RCX and Parasitic Extraction

After successful DRC and LVS verification, RC parasitic extraction was performed using RCX.

Physical interconnections in a real integrated circuit introduce additional resistance and capacitance. These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed OR gate is shown in Figure 4.

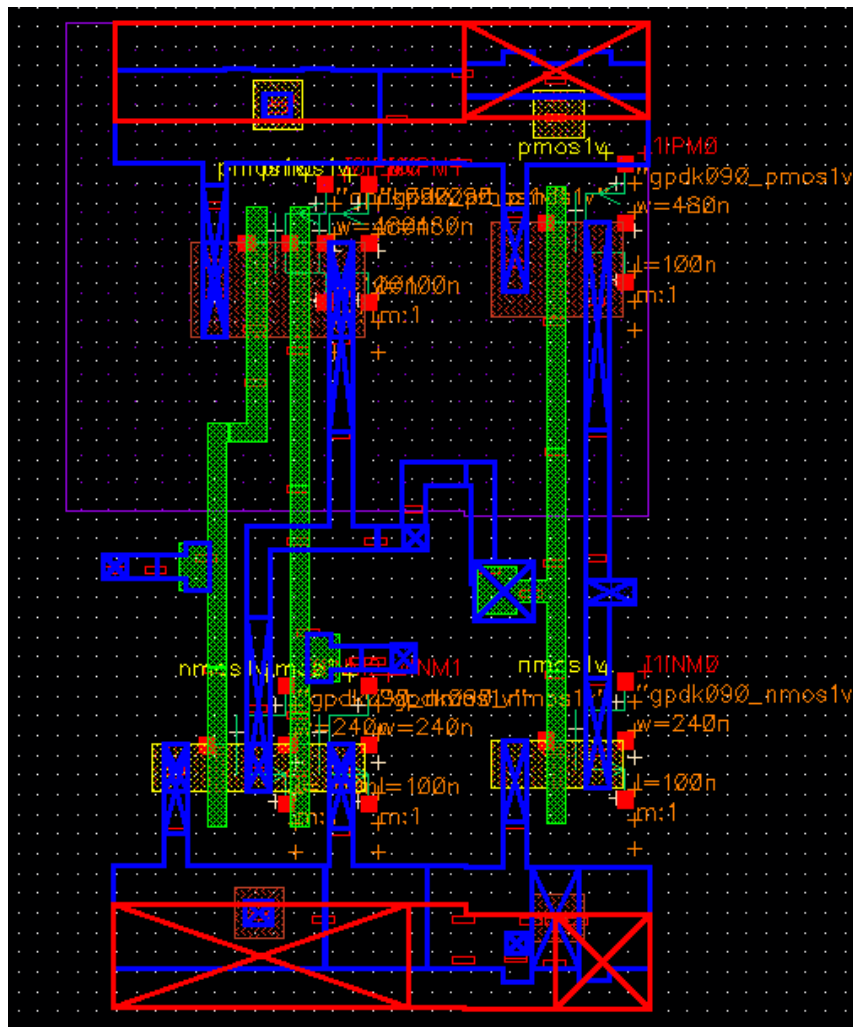


Figure 4: RCX/extracted view of the two-input CMOS OR gate.

The RCX process was successfully completed.

RCX: Successfully Completed

The extracted circuit was used for post-layout simulation and performance analysis.

9 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the extracted view of the OR gate.

Unlike ideal schematic simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the performance of the fabricated circuit.

The main performance parameters calculated in this project were:

1. Average power consumption
2. Propagation delay

10 Average Power Calculation

The average power consumption of the OR gate was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

The average power over a time interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (10)$$

where $p(t)$ represents the instantaneous power consumption.

In Cadence, the average value of the transient power waveform was calculated using the calculator.

The power calculation is shown in Figure 5.

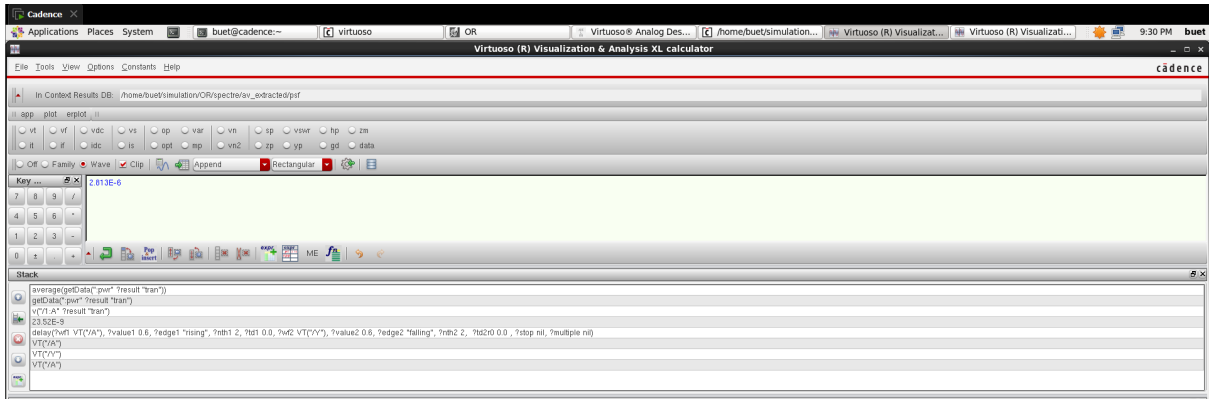


Figure 5: Average power calculation of the extracted OR gate using the Cadence Virtuoso Calculator.

From the Cadence Calculator, the obtained average power is

$$P_{\text{avg}} = 2.813 \times 10^{-6} \text{ W.} \quad (11)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (12)$$

the average power consumption can be written as

$$P_{\text{avg}} = 2.813 \mu\text{W} \quad (13)$$

Therefore, the post-layout average power consumption of the designed two-input CMOS OR gate is approximately

$$2.813 \mu\text{W}.$$

11 Propagation Delay Calculation

Propagation delay represents the amount of time required for an input transition to produce the corresponding transition at the output of a digital circuit.

For a CMOS logic gate, propagation delay is generally measured between the reference voltage crossing of the input signal and the corresponding reference voltage crossing of the output signal.

In this simulation, a switching reference level of approximately 0.6 V was used.

If an input transition occurs at t_{in} and the corresponding output transition occurs at t_{out} , then the propagation delay can be expressed as

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (14)$$

Cadence Virtuoso Calculator was used to determine the delay between the selected transition of input A and the corresponding transition of output Y .

The delay calculation is shown in Figure 6.

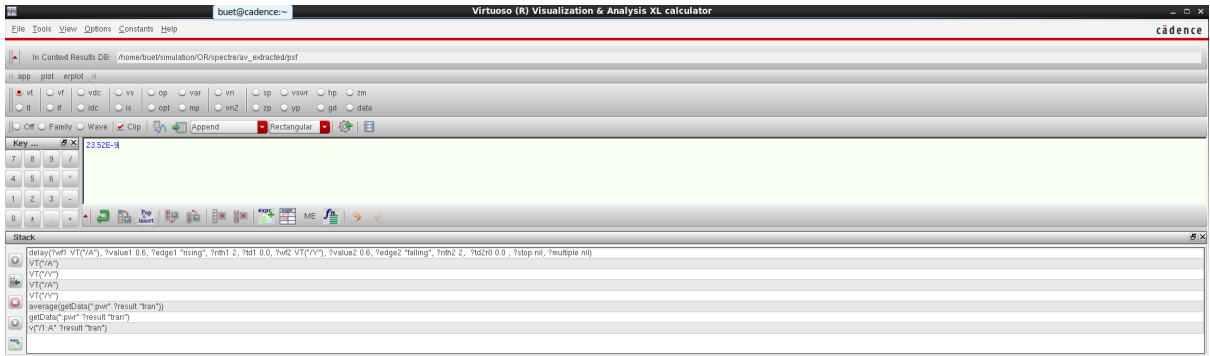


Figure 6: Propagation delay calculation of the extracted OR gate using the Cadence Virtuoso Calculator.

The Cadence Calculator produced a delay of

$$t_p = 23.52 \times 10^{-9} \text{ s.} \quad (15)$$

Since

$$1 \text{ ns} = 10^{-9} \text{ s,} \quad (16)$$

the propagation delay becomes

$$t_p = 23.52 \text{ ns} \quad (17)$$

Therefore, the measured post-layout propagation delay for the selected input-output transition of the CMOS OR gate is approximately

$$23.52 \text{ ns}.$$

12 Verification and Performance Summary

The overall design, verification, and performance results of the two-input CMOS OR gate are summarized in Table 2.

Table 2: Summary of the OR Gate Design Results

Parameter	Result
Logic Function	$Y = A + B$
CMOS Implementation	NOR gate followed by inverter
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	$2.813 \mu\text{W}$
Propagation Delay	23.52 ns

13 Discussion

The complete custom CMOS design flow for a two-input OR gate was successfully carried out using Cadence Virtuoso.

The desired OR operation was implemented using a two-input NOR gate followed by a CMOS inverter. The NOR gate generates the complement of the OR function, while the inverter restores the required logic polarity.

After completing the schematic, a symbol was generated so that the OR gate could be used conveniently as a hierarchical block in larger digital circuits.

The physical layout was then developed using the appropriate transistor, diffusion, polysilicon, metal, contact, well, power, and ground layers.

The layout successfully passed DRC verification. This indicates that the physical dimensions and spacings used in the layout satisfy the fabrication rules of the selected CMOS technology.

The design also successfully passed LVS verification. This confirms that the electrical connectivity of the physical layout matches the original schematic.

After DRC and LVS verification, RCX was performed to extract the parasitic resistance and capacitance associated with the layout. The extracted circuit was then used for post-layout simulation.

The average post-layout power consumption was measured as

$$P_{\text{avg}} = 2.813 \mu\text{W}. \quad (18)$$

The propagation delay for the selected input-output transition was measured as

$$t_p = 23.52 \text{ ns}. \quad (19)$$

These post-layout results provide a more realistic representation of the expected circuit performance because the effects of layout parasitics are included in the simulation.

14 Conclusion

A two-input CMOS OR gate was successfully designed, implemented, and verified using Cadence Virtuoso.

The required logic function was

$$Y = A + B. \quad (20)$$

The OR gate was implemented using a two-input NOR gate followed by a CMOS inverter. The project successfully covered the major steps of a custom VLSI design flow, including:

- schematic design,

- symbol generation,
- physical layout design,
- DRC verification,
- LVS verification,
- RC parasitic extraction, and
- post-layout simulation.

The physical layout successfully passed both DRC and LVS verification, which confirms that the design satisfies the required physical design rules and electrically matches the intended schematic.

RCX was also completed successfully, allowing parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$P_{\text{avg}} = 2.813 \mu\text{W} \quad (21)$$

and the measured propagation delay was

$$t_p = 23.52 \text{ ns} \quad (22)$$

Therefore, the project successfully demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-input CMOS OR gate using Cadence Virtuoso.

15 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator